

Cell Cycle: Copy, Check, Divide

LAUNCH • Science • 40 minutes

I can describe the cell cycle, distinguishing interphase from mitosis and cytokinesis, and sequence prophase, metaphase, anaphase and telophase using chromosome evidence.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

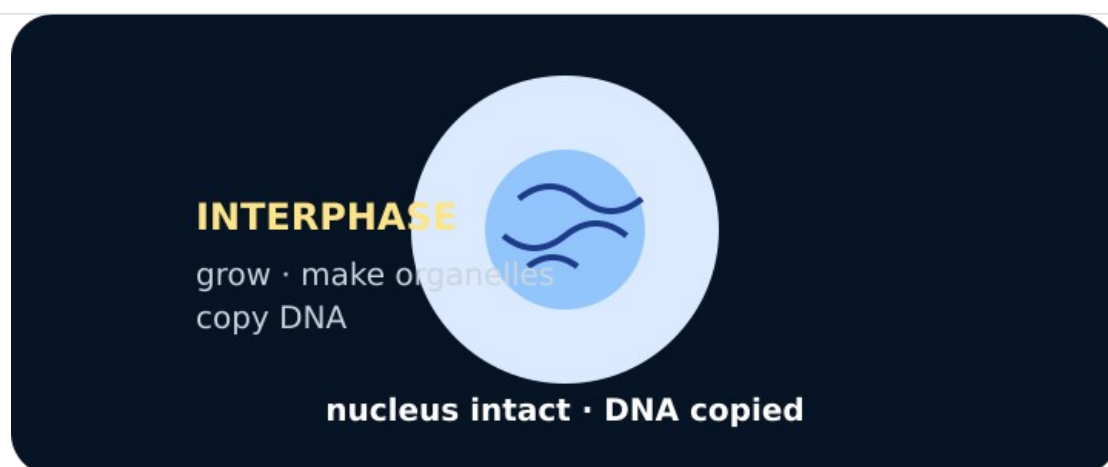
Word	Meaning
interphase	prepare and copy
mitosis	divide the nucleus
cytokinesis	split the cell

First idea

A cell has grown and copied its DNA, but the chromosome copies have not yet separated. Which part of the cycle is it completing?

- A. Interphase, which prepares the cell for nuclear division.
- B. Cytokinesis, because the cytoplasm has already formed two cells.
- C. The end of the cycle, because DNA copying alone makes two daughter cells.

My first idea and reason:



Interphase cell with intact nucleus

We do • choose, explain, check

Which event belongs to interphase rather than mitosis?

- A. Copied chromosomes align across the middle of the cell.
- B. The cell replicates its DNA before nuclear division.
- C. Sister chromosome copies move to opposite ends of the cell.

My choice: _____ Evidence or reason:

Cell-cycle event routing console

Route each event to preparation before nuclear division, nuclear division, or cell split/outcome, then build one defensible sequence.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
INTERPHASE • PREPARE	
MITOSIS • NUCLEAR DIVISION	
CYTOKINESIS • CELL SPLIT	

Activity cards

1 • Cell growth The cell grows and increases the number of subcellular structures.	2 • DNA replication Each chromosome's DNA is copied before nuclear division.
3 • Chromosomes condense Copied chromosomes become shorter, thicker and more visible in prophase.	4 • Chromosomes align Copied chromosomes arrange near the equator of the cell in metaphase.
5 • Sister chromatids separate Copied chromosome halves move towards opposite poles in anaphase.	6 • Two nuclei reform Nuclear envelopes form around the two chromosome groups during telophase.

7 • Cytoplasm divides

The membrane and cytoplasm separate the parent cell into two daughter cells.

Describe the complete cycle using interphase, one visible clue for each mitotic stage, and cytokinesis.

Your lesson record

Create an evidence-labelled cell-cycle sequence and answer GCSE DESCRIBE and EXPLAIN questions that distinguish interphase, mitosis and cytokinesis.

Model helps us see

Use the chain PREPARE → ALIGN → SEPARATE → REFORM → SPLIT, then replace each everyday word with the formal stage and its chromosome evidence.

Model does not show

Stage cards freeze a continuous, three-dimensional process into neat snapshots. Chromosomes, spindle fibres, cell size and timing are not to scale, and checkpoints and molecular controls are mostly omitted.

Supported

Order six labelled picture cards, match prepare/divide/split, and complete a stem explaining when DNA is copied.

Use: During interphase ___. During mitosis ___. During cytokinesis ___.

Standard

Sequence interphase, four named mitotic stages and cytokinesis, annotate visible evidence, then explain why DNA replication must precede mitosis.

Use DESCRIBE for the ordered changes and EXPLAIN for cause plus biological reason.

Stretch

Evaluate an incorrect cell-cycle model, repair chromosome-number reasoning and explain why stage boundaries can be uncertain in a snapshot.

Distinguish observation, stage inference and model limitation; include the caveat that mutation or segregation error can disrupt the usual outcome.

Evidence target

A complete ordered cycle, at least four stage-evidence links, a correct distinction between DNA replication and chromosome separation, and one pupil-owned explanation.

Exit, Audience and evidence • W9L1

Supported: Complete: DNA is copied during ___, chromosome copies separate during ___, and the cytoplasm divides during ___.

Standard: Describe the cell cycle from interphase to two daughter cells using at least four formal stage terms.

Stretch: Explain why DNA amount increases before division without each daughter cell retaining twice the parent chromosome number.

What I said, and what it changed

Next I will _____.

Adult witness note

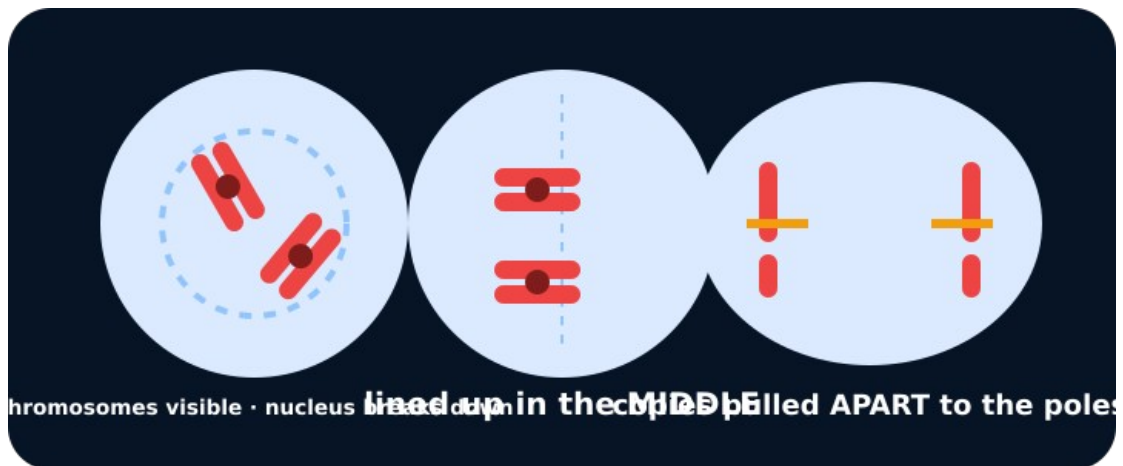
What was observed, and with what level of independence?

My evidence and explanation

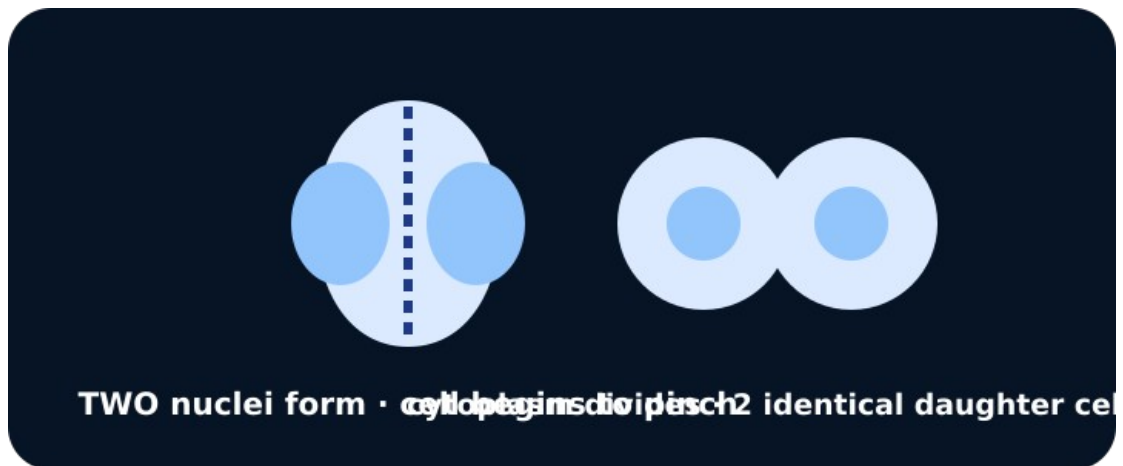
What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



Prophase, metaphase, anaphase



Telophase then cytokinesis

Exit • one next step

After the adult has listened, what will you keep, improve or check next?